METRIC
MIL-DTL-32532A
4 August 2021
SUPERSEDING
MIL-DTL-0032532(SH)
18 September 2015

DETAIL SPECIFICATION

# LIFE RAFT, INFLATABLE, 25-PERSON (MARK-7) MOD 2, AND 50-PERSON (MARK-8) MOD 2

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the 25-person (Mark-7) Mod 2 and 50-Person (Mark-8) Mod 2 inflatable life rafts, including inflation system and survival gear, packed within a rigid stowage container, ready for use as abandon ship survival equipment.

## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

| | | |
|---|---|---|
| MIL-R-52255 | - | Repair Kit and Repair Kit Components, for Collapsible Fabric Tanks, Drums, and Boats |
| MS27826 | - | Plug Assembly (Sealing Clamp) Life Raft Repair |

DEPARTMENT OF DEFENSE STANDARDS

| | | |
|---|---|---|
| MIL-STD-810 | - | Environmental Engineering Considerations and Laboratory Tests |

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 4220

DISTRIBUTION STATEMENT A. Approved for public release. Distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:52Z
Check the source to verify that this is the current version before use.

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

| | | |
|---|---|---|
| 46 CFR 160.026 | - | Water, Emergency Drinking (in Hermetically Sealed Containers), for Merchant Vessels |
| 46 CFR 160.043 | - | Jackknife (with Can Opener) for Merchant Vessels |
| 46 CFR 160.054 | - | Kits, First-Aid, for Inflatable Liferafts |
| 46 CFR 160.061 | - | Fishing Tackle Kits, Emergency, for Merchant Vessels |
| 46 CFR 160.151 | - | Inflatable Liferafts (SOLAS) |
| 46 CFR 164.018 | - | Retroreflective Material for Lifesaving Equipment |

(Copies of these documents are available online at www.ecfr.gov.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

| | | |
|---|---|---|
| 583-5106577 | - | Container - Type II - Fiberglass Reinforced Plastic for 50 Person Inflatable Life Raft |
| 583-5111055 | - | Container, U.S.N. Life Raft, 25-Person |

(Copies of these documents are available online at https://199.208.213.105/webjedmics/index.jsp. To request an NSEDR account for drawing access, send an email to NNSY_JEDMICS_NSEDR_HELP_DESK@navy.mil.)

NAVSEA TECHNICAL PUBLICATIONS

| | | |
|---|---|---|
| S9008-EA-INS-010 | - | Survival at Sea Instructions for the Operation and Use of the U.S. Navy 25-Person Life Raft (Mark 7) and 50-Person Life Raft (Mark 8) |

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the TMIN without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library at https://nll.ahf.nmci.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

| | | |
|---|---|---|
| ASTM B36/B36M | - | Standard Specification for Brass Plate, Sheet, Strip, And Rolled Bar |
| ASTM D413 | - | Standard Test Methods for Rubber Property–Adhesion to Flexible Substrate |
| ASTM D471 | - | Standard Test Method for Rubber Property–Effect of Liquids |
| ASTM D751 | - | Standard Test Methods for Coated Fabrics |
| ASTM D2261 | - | Standard Test Method for Tearing Strength of Fabrics by the Tongue (Single Rip) Procedure (Constant-Rate-of-Extension Tensile Testing Machine) |
| ASTM D3787 | - | Standard Test Method for Bursting Strength of Textiles—Constant-Rate-of-Traverse (CRT) Ball Burst Test |



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:52Z
Check the source to verify that this is the current version before use.

ASTM D3884 - Standard Guide for Abrasion Resistance of Textile Fabrics (Rotary Platform, Double-Head Method)

ASTM D5035 - Standard Test Method for Breaking Force and Elongation of Textile Fabrics (Strip Method)

(Copies of these documents are available online at www.astm.org.)

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC TM169 - Weather Resistance of Textiles: Xenon Lamp Exposure

(Copies of this document are available online at https://aatcc.org.)

COMPRESSED GAS ASSOCIATION, INC. (CGA)

CGA G-7.1 - Commodity Specification for Air

(Copies of this document are available online at www.cganet.com.)

DEFENSE ACQUISITION UNIVERSITY (DAU)

Nomograph of Cumulative Binomial Distribution – Sampling Chart

(Copies of this document are available online at https://myclass.dau.edu/bbcswebdav/xid-2521820_4.)

INTERNATIONAL MARITIME ORGANIZATION (IMO)

IMO Resolution MSC.48(66) Chapter I, Regulation 1.2 - General Requirements for Life-Saving Appliances

IMO Resolution MSC.48(66) Chapter IV, Regulation 4.1 - General Requirements for Liferafts

IMO Resolution MSC.48(66) Chapter IV, Regulation 4.2 - Inflatable Liferafts

IMO Resolution MSC.81(70) - Revised Recommendation on Testing of Life-Saving Appliances

SOLAS V/16 - Safety of Navigation – Maintenance of Equipment

(Copies of these documents are available online at www.imo.org.)

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY (NIST)

NIST SP 440 - Color: Universal Language and Dictionary of Names

NIST SP 1038 - The International System of Units (SI) - Conversion Factors for General Use

(Copies of these documents are available online at https://archive.org/details/NISTresearchlibrary.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:52Z
Check the source to verify that this is the current version before use.

## 3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Regulatory compliance. The life raft shall be an existing product or non-developmental item (NDI) that, as of the date of the solicitation, holds U.S. Coast Guard (USCG) approval. Except as specified herein, the inflatable life raft shall also meet the requirements of Chapter IV of MSC.48(66) and 46 CFR 160.151. The inflatable life raft shall be constructed in accordance with the following the International Convention for the Safety of Life at Sea (SOLAS) 74/83, hereinafter referred to as SOLAS, provisions:

a. IMO Resolution MSC.48(66) Chapter I, Regulation 1.2

b. IMO Resolution MSC.48(66) Chapter IV, Regulation 4.1

c. IMO Resolution MSC.48(66) Chapter IV, Regulation 4.2

3.4 Design and construction.

3.4.1 Shape. The inflated life raft may be circular, hexagonal, octagonal, elliptical, or similar outline in plan view.

3.4.2 Resistance to deterioration. All materials used in the construction of the life raft shall not be subject to deterioration from the effects of weathering aboard ship under the conditions of usual stowage, nor from contact with salt water or spray or petroleum products. The life raft shall be resistant to deterioration where exposed to sunlight.

3.4.3 Stability. The life raft shall have water-capturing stability appendages on its underside, to resist capsizing from wind and wave forces. These appendages shall meet the following requirements:

a. The total volume of the appendages shall be not less than 500 liters for the Mark-7 life raft and 1,000 liters for the Mark-8 life raft. Any opening designed to close as water is forced out of an appendage is not considered an opening for the purpose of this calculation.

b. The appendages shall be securely attached and evenly distributed around the periphery of the exterior bottom of the life raft. The appendages may be omitted at the locations of inflation cylinders.

c. Openings in or between the stability appendages shall be provided to limit the formation of air pockets under the life raft.

d. The appendages shall be designed to deploy underwater when the life raft inflates. If weights are used for this purpose, they shall be of non-corrodible material or protected against corrosion.

3.4.4 Body.

3.4.4.1 Buoyancy chambers. Two main buoyancy chambers shall be superimposed, one atop the other, each inflated through a non-return inflation valve on each chamber. The buoyancy chambers shall be arranged so that, in the event of any one of the chambers being damaged or failing to inflate, the intact compartments shall be able to support, with positive freeboard over the life raft's entire periphery, 25 persons (Mark-7) or 50 persons (Mark-8), each having a mass of 82.5 kilograms and seated in their normal positions.

3.4.4.2 Storage pockets. Pockets or equivalent stowage bags shall be fitted for stowage of hand pumps, repair kits, instruction manuals, and other equipment needed immediately on launching.

3.4.4.3 Reinforcing patches. Reinforcing patches shall be substantially fitted in way of the attachments for the righting line, painter, sea anchor line, etc.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:52Z
Check the source to verify that this is the current version before use.

3.4.4.4 Inflation cylinder holders. The holders for retaining the inflation cylinders shall be substantially fitted and sufficiently strong to retain the cylinders when the life raft is inflated or when the life raft is dropped into the water.

3.4.4.5 Exposed surfaces. All exposed surfaces shall be reasonably smooth and free from sharp protrusions or projections that might be injurious in boarding or occupancy.

3.4.5 Floor. The floor of the life raft shall be waterproof and shall be capable of being sufficiently insulated against cold by means of one or more compartments, which the occupants can manually inflate or which inflate automatically and can be deflated and re-inflated by the occupants. Insulation by other equally efficient means not dependent on inflation may also be used.

3.4.6 Canopy. The life raft design shall incorporate an inner and outer canopy to insulate the occupants from the weather, which shall automatically erect upon initial inflation of the life raft. Means shall be provided to prevent accumulation of water in the air gap. The canopy shall provide headroom for sitting occupants (of average height, i.e., 175.4 centimeters [approximately 5 feet 9 inches]) under all parts of the canopy, and provision shall be made for furling the ends to allow entry of occupants, free passage of air, and room for paddling. Provision for catching rain water shall be incorporated onto the canopy. The canopy shall be provided with at least one viewing port for the Mark-7 and two viewing ports for the Mark-8. Ventilation ports (vents) shall be incorporated in the canopy to allow for air exchanges when canopy doors/entrances are closed (see 3.11.10 and 4.7.8). The interior of the canopy shall be of a color that does not cause discomfort to the occupants.

3.4.7 Entrances. The life raft shall have at least two diametrically opposite entrances. Each entrance shall be clearly indicated and be provided with efficient adjustable closing arrangements which can be easily and quickly opened from inside and outside the life raft so as to permit ventilation, but exclude seawater, wind, and cold.

3.4.8 Laps and seams. The rafts shall be constructed with heat sealed seams. Each seam shall be at least as strong as the weakest of the materials joined by the seam. Each seam shall be covered with tape where necessary to prevent lifting of and damage to fabric edges. Vulcanization shall not be used. Chemically cured adhesives and radio frequency (RF) welded (polyurethane-coated fabrics) methods are acceptable and shall meet the requirements of 3.10.3 and 3.10.4. There shall be no sewing that penetrates the fabric of an air retaining chamber.

3.4.9 Dissimilar materials. Where dissimilar materials are joined in the construction of the inflatable life raft, provisions shall be made to prevent loosening or tightening due to differences in thermal expansion, freezing, buckling, galvanic corrosion, or other incompatibilities.

3.4.10 Retro-reflective material. The life raft shall be marked with type I retro-reflective material approved under 46 CFR 164.018. Retro-reflective materials shall be fitted around the canopy of the life raft. The material shall be sufficiently wide and long to give a minimum area of 150 square centimeters ($cm^2$) and shall be spaced at intervals approximately 80 millimeters from center to center, at a height above the waterline, doorways included, for most visibility. Retro-reflective materials shall also be fitted to the underside of the floor, cross-shaped in the center. The dimension of the cross shall be half the diameter of the life raft and a similar cross shall be applied to the top of the canopy.

3.4.11 Color. The primary color of the exterior of the canopy shall be vivid reddish orange (color number 34 of NIST SP 440), or a fluorescent color of a similar hue. The remainder of the life raft may be the natural color of the finished coated fabric, except that the underside of the floor shall be dark blue, dark grey, or black. In addition, fabric for storage pockets or equivalent stowage bags, including survival equipment bags (see 3.4.4.2 and 3.7.25), shall not be yellow.

3.4.12 Container. The life raft containers shall be constructed in accordance with NAVSEA Drawing 583-5111055 for the Mark-7 life raft and NAVSEA Drawing 583-5106577 for the Mark-8 life raft. However, for both life raft containers, the contractor may determine fiberglass lay-up that uses best manufacturing practices and meets the performance requirements of 3.11.4. The container shall be held closed with ASTM B36/B36M brass bands, 1.9 centimeters (¾ inch) wide, 0.05 centimeter (0.020 inch) thick, which are secured closed with two corrosion-resistant steel (CRES) securing clips on each band; there shall be two bands on the Mark-7 container and four bands on the Mark-8 container. The life raft shall be packed in its container in such a way as to ensure that the waterborne life raft inflates in an upright position on breaking free from its container.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:52Z
Check the source to verify that this is the current version before use.

3.4.13 Inflation. Inflation shall take place upon the pulling of the painter line, which may be accomplished manually both from the deck of a vessel or by a swimmer in the water. In addition, inflation connections shall be provided for use with the manually operated pump or bellows. Refer to 3.6 for the inflation system requirements.

3.4.14 Workmanship and materials. The life raft shall be constructed with proper workmanship and materials. Repairs or rework shall be as specified (see 6.2). The ends of all ropes and webbing shall be fused, whipped, or otherwise secured to prevent fraying. Life raft coated fabric materials shall meet the performance requirements of 3.10.

3.4.15 Temperature ranges. The life raft shall not be damaged in stowage throughout an air temperature range of -30 to +65 °C. The life raft shall operate throughout a seawater temperature range of -1 to +30 °C.

3.5 Life raft fittings. The life raft shall be provided with the following fittings.

3.5.1 Boarding ramp. On the Mark-7 life raft, at least one entrance shall be fitted with a semi-rigid inflatable boarding ramp. The Mark-8 shall have at least two entrances fitted with boarding ramps. The boarding ramp shall have sufficient size and buoyancy to support one person weighing 100 kilograms in a sitting or kneeling position without holding onto any other part of the life raft. The boarding ramp shall be arranged so as to prevent deflation of the life raft if the ramp is damaged.

3.5.2 Boarding ladder. Entrances not provided with a boarding ramp shall have a boarding ladder, the lowest step of which shall be situated not less than 400 millimeters below the life raft's light waterline. The steps of the boarding ladder shall be of rigid or semi-rigid construction and secured against rotation, to provide a firm foothold capable of supporting a person weighing 100 kilograms, at a minimum. There shall be means inside the life raft to assist persons in pulling themselves into the life raft from the ladder.

3.5.3 Exterior canopy lamp. The Mark-7 life raft shall be equipped with one exterior canopy lamp; the Mark-8 life raft shall have two. The lamp shall be the exterior raft light, part number 47800, from Adventure Lights Inc., or equal (see 6.12.1).

3.5.4 Interior canopy lamp. The Mark-7 life raft shall have one interior canopy lamp, fitted inside the life raft; the Mark-8 life raft shall have two. The lamp shall the interior/boarding raft light, part number 47850, from Adventure Lights Inc., or equal (see 6.12.2).

3.5.5 Lifelines. One lifeline shall be fitted around the outside periphery and one around the inside of the life raft. The two lifelines shall be made of 14-millimeter minimum width nylon tubular webbing, or 10-millimeter minimum diameter rope, or equivalent. Each lifeline attachment patch shall have a minimum breaking strength of 1.5 kiloNewton (kN) pull exerted in a direction perpendicular to the base of the patch. Each bight of the exterior lifeline shall be long enough to allow the lifeline to reach to the waterline of the life raft when it is afloat.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:52Z
Check the source to verify that this is the current version before use.

3.5.6 Sea painter system. A sea painter with bung assembly shall be provided for inflating and holding the life raft during boarding. It shall be a sea painter with bung assembly, part number HRC-950, from Hampton Rubber Co., or equal (see 6.12.3). The sea painter, with an approximate breaking strength of 1.77 kN, shall act as the weak link. The sea painter and rubber plug shall be installed as shown on figure 1.

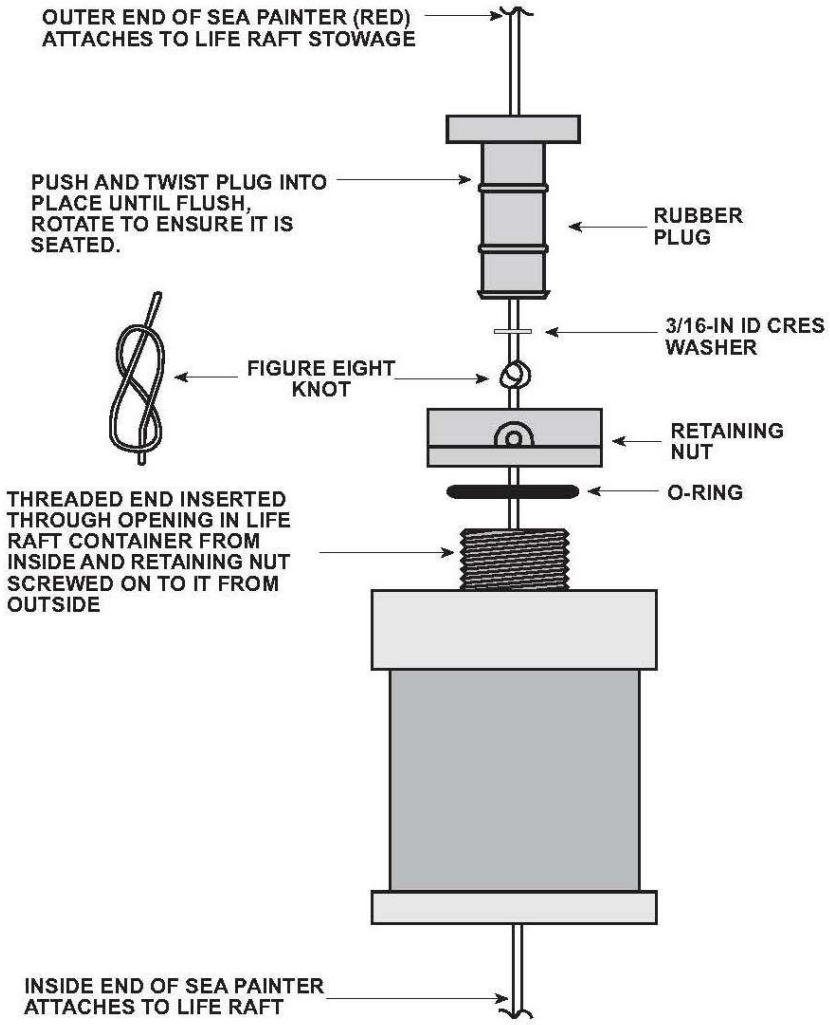



FIGURE 1. Sea painter/bung assembly arrangement.

3.5.7 Righting gear. Handholds or webbing straps on the underside of the floor shall be provided to enable a righting moment to be exerted. Handholds or webbing straps shall be of sufficient strength to allow for righting of the inverted life raft. The righting strap shall run the full width of the life raft and shall be secured to the buoyancy chamber on both sides of the life raft. The stability of the life raft when in the inverted position shall be such that it can be righted in a sea way and in calm water by one person.

3.5.8 Towing/mooring connections. Four towing/mooring connections shall be provided, one each at opposite ends of the life raft on the lower inflatable hull tube section and two connections spaced equidistant on the lower hull tube between those two fittings. The mooring/towing connections shall be marked to indicate their function. The life raft and its fittings shall be constructed so as to enable it to be towed at a speed of 3 knots in calm water when loaded with its full complement of persons and equipment and with one of its sea anchors deployed. See 3.7.24 for associated hook requirements.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:52Z
Check the source to verify that this is the current version before use.

3.6 Inflation system. The life raft shall be equipped with an automatic inflation system, which consists of a commercial off-the-shelf (COTS) standard Department of Transportation (DOT)-approved carbon fiber over aluminum composite cylinder, a shop repairable inflation valve, and a multi-use inflation hose with end fittings compatible with the inflation valve and the raft inlet check valve. The inflation gas shall be Grade D breathable air in accordance with CGA G-7.1, with a maximum pressure of 31 megapascals (MPa) at 21 °C. The inflation valve shall be serviceable by life raft shop personnel with an appropriate rebuild kit available.

3.6.1 Inflation cylinders. Inflation cylinders shall be carbon fiber wrapped aluminum composite air cylinders with a minimum service life of 15 years. The Mark-7 life raft inflation cylinders shall have a female valve port sized to accept an inflation valve with 0.875-14 UNF (male) threads and an O-ring mating surface such that a minimum of six threads are engaged and there is a maximum working pressure of 31 MPa (4,500 pounds per square inch [psi]). The Mark-7 cylinder shall have a volume of 9,013 centimeters cubed ($cm^3$) (550 cubic inches [$in^3$]), a diameter of 17.78±0.635 centimeters (7±0.25 inches), and an overall length of 54.61±1.27 centimeters (21.5±0.5 inches). The Mark-8 composite inflation cylinders shall be sized as necessary to meet inflation requirements of the life raft in all stated temperature extremes. Each compressed gas inflation cylinder within the life raft shall be installed as follows:

a. The slings and reinforcements shall be of sufficient strength to retain the inflation cylinders in place when the life raft is dropped into the water from a height of 23 meters during deployment.

b. The painter and inflation cylinders of the life raft are linked to start inflation when the painter is pulled by one person exerting a force not exceeding 250 Newtons (N).

3.6.2 Inflation valve. The inflation valve shall be equipped with a safety relief feature which will release the cylinder pressure when the cylinder reaches a pressure of five-thirds the maximum working pressure of the inflation cylinder.

3.6.3 Pressure relief valve poppet retainer. Poppet retainers for rendering pressure relief valves inoperable shall be provided for every pressure relief valve installed in the life raft, unless the pressure relief valves are of a type that can be rendered inoperable. If provided, pressure relief valve poppet retainers shall either float or be secured to the life raft by a lanyard.

3.6.4 Valves. The pressure relief valves, inlet check valves, and topping off/dumping valves shall fit in a hull tube-mounted flange system with a 1.312016 UN-2B thread opening. The flanges shall be mounted in the hull tubes, arch tube, and any other inflatable section requiring these types of valves. The flanges shall be adhered to the inflatable sections using RF welding for polyurethane-coated fabrics or appropriate adhesive for the materials being joined. The relief valves shall open/relieve at a pressure not greater than 130 percent of the designed working pressure, and shall close/reseat at a pressure not less than 100 percent of the designed working pressure. There shall be a minimum differential between opening and closing pressures of 21 millibar (0.3 psi). The relief valves shall be as small as practicable, and shall be mounted on a flat molded universal flange that can be easily attached to the life raft flotation tubes. All metal parts that can cause damage to the life raft fabric shall be covered with molded neoprene or polyurethane.

3.7 Life raft equipment. The equipment in paragraphs 3.7.1 through 3.7.25 shall be provided with each life raft.

3.7.1 Buoyant rescue quoit and heaving line. A buoyant rescue quoit and heaving line not less than 30 meters in length and having a breaking strength of not less than 1.1 kN shall be provided and shall be attached to the life raft near the entrance furthest from the painter attachment.

3.7.2 Knife. Two knives shall be provided. One knife of the non-folding type having a buoyant handle and lanyard attached shall be stowed in a pocket on the exterior of the canopy near the point at which the painter is attached, and one of the folding type in accordance with 46 CFR 160.043 shall be stowed in the survival equipment bag or bags.

3.7.3 Bailer. One or two bailers having a total aggregate volume of at least 2 liters shall be provided. Each bailer shall be equipped with a lanyard which is at least 1 meter in length.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:52Z
Check the source to verify that this is the current version before use.

3.7.4 Cellulose sponge. Two cellulose sponges, each having a volume of at least 750 $cm^3$ when saturated with water, shall be provided.

3.7.5 Sea anchor. Two sea anchors, each with a shock-resistant hawser and tripping line, shall be provided. One sea anchor, being the primary sea anchor, shall be permanently attached to the life raft in such a way that, when the life raft inflates or is waterborne, it will cause the life raft to lie oriented to the wind in the most stable manner, and the other, being a spare sea anchor, shall be packed in the survival equipment. The strength of each sea anchor and its hawser and tripping line shall be selected for all sea conditions. The sea anchors shall be fitted with a swivel at each end of the line and shall be of a type that is unlikely to turn inside-out between its shroud lines. Sea anchors without the swivel may be used if, during the towing test, their design is demonstrated to be of a type that does not rotate when streamed. The sea anchors are not required to have the tripping lines if, during the towing test, their design is demonstrated to be of a type that can be hauled in by one person.

3.7.6 Buoyant paddles. Two buoyant paddles at least 1.2 meters long shall be provided. The paddles shall be of the same size and type as used to pass the maneuverability test of Part 1, 5.10 of IMO Resolution MSC.81(70).

3.7.7 Pump or bellows. A manually operated pump or bellows, with a connecting fitting compatible with the topping-off valve and capable of inflating any inflatable chamber to its working pressure, shall be provided. The Mark-7 life raft shall have at least one pump or bellows and the Mark-8 shall have at least two pumps or bellows.

3.7.8 First-aid kit. The Mark-7 life raft shall be provided with one First Aid Kit (FAK); the Mark-8 life raft shall have two. The FAK shall be the "TacMed™ Life Raft FAK", product number 80-0034, from Tactical Medical (TacMed) Solutions or equal (see 6.12.4). All contents of the FAK shall have a minimum shelf-life of 12 months remaining, when the FAK is packed in the life raft container.

3.7.9 Abandon ship signal kit. An abandon ship signal kit (National Stock Number [NSN] 1370-01-366-0344 or 1370-01-418-2657) shall be packed with the survival equipment. The kit shall contain six Mark-124 day/night signals and two Mark-79 aerial signal kits. The abandon ship signal kit shall be Government-furnished property (GFP), as specified (see 6.2 and 6.3).

3.7.10 Electric flashlight. One electric LED flashlight shall be provided. It shall be capable of a minimum of 60 lumen output after 4 hours of use, be bright orange or yellow in color, and shall use no more than 4 AA 1.5 VDC lithium batteries. The maximum brightness shall not exceed 120 lumens. It shall be waterproof to a depth of 2 meters. It shall have an on/off switch that allows for momentary signaling, such as Morse code signaling. It shall have a lanyard with a circumferential length of 1 meter±5 centimeters to hang about the neck.

3.7.11 Mirror, signaling. One daylight signaling mirror with instructions on its use for signaling to ships and aircraft shall be provided.

3.7.12 Whistle. One ball type whistle or multi-tone whistle, made of corrosion-resistant materials, shall be provided.

3.7.13 Lifesaving signal waterproof card. One copy of the lifesaving signals referred to in SOLAS V/16 shall be provided on a waterproof card or in a waterproof container.

3.7.14 Fishing tackle. One emergency fishing tackle kit shall be provided in accordance with 46 CFR 160.061.

3.7.15 Food rations. Each Mark-7 life raft shall be equipped with 25 food rations and each Mark-8 with 50 food rations. Each food ration shall total not less than 10,000 kilojoules. The rations shall be kept in airtight packaging and shall be stowed in a watertight container. Rations shall be palatable and have a minimum shelf-life of 5 years and a minimum remaining shelf-life of 48 months when packed in the life raft container. Each food ration shall be packaged in a manner whereby it can be readily divided into daily portions and easily opened. Each ration shall have its contents packed into a minimum of six individually wrapped portions. Each ration shall be labeled "Ration for one person for three days" and shall have a USCG approval number.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:52Z
Check the source to verify that this is the current version before use.

3.7.16 Drinking water. Each Mark-7 life raft shall be packed with 25 bottles of drinking water and each Mark-8 with 50 bottles of drinking water. Each 500-milliliter plastic watertight bottle of potable water shall meet the requirements of 46 CFR 160.026. The bottle shall be graduated in 100-milliliter increments. The drinking water shall have a minimum overall shelf-life of 5 years and shall be packed in the life raft container with not less than 48 months remaining of the shelf-life.

3.7.17 Manual reverse osmosis desalinator (MROD). One MROD shall be packed with the survival equipment. The MROD (NSN 4610-01-319-5337) shall be GFP, as specified (see 6.2 and 6.3).

3.7.18 Anti-seasickness medicine. Six doses of anti-seasickness medicine shall be provided for each person (Mark-7: 25 persons, Mark-8: 50 persons). There shall be a minimum of 12 months remaining of the shelf-life for the anti-sickness medicine when packed in the life raft container. The medication shall be one of the following:

a. Dimenhydrinate in 50-milligram tablets;

b. A combination of 25 milligrams of promethazine hydrochloride and 25 milligrams of ephedrine sulfate, comprising each single dose, to be taken at 6-hour intervals; or

c. A transdermal patch containing scopolamine formulated for at least 2 days' use, with each patch counted as six doses.

3.7.19 Survival instructions. Each life raft shall be equipped with one waterproof hardcopy of S9008-EA-INS-010 (NSN 0910-LP-102-2272), which are the instructions on how to survive in a life raft. The instructions will be provided as GFP (see 6.2 and 6.3), and shall be suspended in a clear film envelope from one of the canopy arch tubes.

3.7.20 Immediate action instructions. Each life raft shall be equipped with one hardcopy of immediate action instructions. The instructions for immediate action shall:

a. Be waterproof.

b. Be in English.

c. Explain both the noise accompanying the operation of any provided pressure relief valves and the need to render them inoperable after they complete venting.

d. Be suspended from the inside canopy, so they are immediately visible by survivors upon entering the life raft. They may be contained in the same envelope with the instructions on how to survive, if the instructions for immediate action are visible from both faces of the envelope.

3.7.21 Thermal protective aid. Thermal protection shall be provided for the occupants of the life raft by aluminized Mylar thermal protective blankets. The Mark-7 shall be equipped with 12 each and the Mark-8 with 25 each thermal protective blankets. The blankets shall have the following salient characteristics:

a. Be made of waterproof material having a thermal conductivity of not more than 0.25 watt per square meter degree Kelvin.

b. Have a physical size of 2.15 by 1.4 meters, with a tolerance in both length and width of ±0.15 meter.

c. Have coated material with a minimum tensile strength of 82.7 Newtons per square millimeter ($N/mm^2$) and a thickness of 0.012 millimeter, -0.0 millimeter.

d. Be packed in a waterproof package with a desiccant pack.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:52Z
Check the source to verify that this is the current version before use.

3.7.22 Repair outfit. Repair kits shall be provided and installed in a pocket or survival equipment bag inside the life raft. The Mark-7 shall have two repair kits and the Mark-8 shall have four repair kits. Each kit shall:

a. Be sealed in a waterproof package with a desiccant pack to prevent moisture.

b. Contain one each of the size sealing clamps specified in table I in accordance with MS27826 or per the guidance in MIL-R-52255 (see 6.12.6).

TABLE I. Clamp size guidance chart.

| | **MS27826** | | **MIL-R-52255** | |
|---|---|---|---|---|
| CLAMP SIZE (Approximated) | PART NO. | NSN | PART NO. | NSN |
| 76-millimeter (3-inch) | MS27826-1 | 5340-00-720-8864 | N/A | 5342-00-720-8864 |
| 127-millimeter (5-inch) | MS27826-2 | 5340-00-720-8863 | N/A | 5342-00-720-8863 |
| 190-millimeter (7½-inch) | MS27826-3 | 5340-00-720-8858 | N/A | 5342-00-720-8858 |

3.7.23 Spare battery pack. Each life raft shall have a spare battery pack located in the same survival equipment bag where the flashlight is stowed. The spare battery pack shall be configured so that the terminals on the batteries cannot make contact with the terminals of other batteries within the package. The batteries shall be stored in a re-closeable waterproof pouch or container along with a desiccant pack. The spare battery pack for the Mark-7 life raft shall contain 10 each 1.5-volt DC, AA-sized lithium batteries, and the spare battery pack for the Mark-8 life raft shall contain 14 each 1.5-volt DC, AA-sized lithium batteries.

3.7.24 Towing/mooring connection hooks. Two hooks shall be provided for life raft occupants to easily attach to the towing/mooring connections specified in 3.5.8 and to facilitate connecting the life raft to other life rafts when afloat. The hooks shall be part #2337, “15/32 STD Carbine W/O Eye”, from Wichard America, or equal (see 6.12.5).

3.7.25 Survival equipment bag(s). The life raft shall be provided with a survival equipment stowage bag or bags of a waterproof material having a re-sealable, watertight opening; these shall be capable of being tied to the inside of the life raft with short lengths of nylon line, webbing, or equivalent and shall be sized to stow the below-listed equipment. The survival equipment bag(s) shall be located and secured inside the life raft and shall remain inside the life raft upon inflation.

a. Knife, folding (see 3.7.2)
b. Bailer(s) (see 3.7.3)
c. Sponges (see 3.7.4)
d. FAK (see 3.7.8)
e. Abandon ship signal kit (see 3.7.9)
f. Electric flashlight and batteries (see 3.7.10)
g. Mirror, signaling (see 3.7.11)
h. Whistle (see 3.7.12)
i. Fishing tackle (see 3.7.14)
j. Food rations (see 3.7.15)
k. Drinking water, bottled (see 3.7.16)
l. MROD (see 3.7.17)
m. Anti-seasickness medicine (see 3.7.18)



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:52Z
Check the source to verify that this is the current version before use.

n. Thermal protective aids (see 3.7.21)

o. Spare battery pack (see 3.7.23)

p. Towing/mooring connection hooks (see 3.7.24)

3.8 Size and mass. The mass of the Mark-7 life raft, including its container and equipment, shall not exceed 185 kilograms (408 pounds). The mass of the Mark-8 life raft, including its container and equipment, shall not exceed 340 kilograms (750 pounds).

3.9 Marking.

3.9.1 Life raft nameplate. The life raft shall have a permanently attached substantial nameplate of compatible material that is embossed or imprinted with the following:

a. Life Raft, Inflatable

b. Mark-7 Mod 2, 25 Persons or Mark-8 Mod 2, 50 Persons (as applicable)

c. NSN

d. Manufacturer Name and CAGE Code

e. Date of Manufacture (month, year)

f. Contract No. _____

g. Serial No. _____

3.9.2 Life raft container data plate. The life raft container shall have a non-ferrous metal data plate affixed to the top half of the container, with compatible non-corrosive pop rivets. The data plate shall include the following information, with a minimum text height of 3 millimeters:

a. “U.S. NAVY LIFE RAFT”

b. “MK-7 MOD 2, 25 PERSONS” or “MK-8 MOD 2, 50 PERSONS” (as applicable)

c. “MFR” (plus name of manufacturer)

d. “S/N” (plus serial number)

e. “CERT DATE” (plus date of certification)

f. “EXP DATE” (plus date of certification expiration)

g. “FLARE LOT NUMBER” (plus lot number from GFP signal kit)

h. Certification Facility

3.9.3 Serial numbers. A ten-character alpha-numeric serial number shall be assigned to each life raft to provide the life raft with a unique and systematic serial number regardless of contracts or purchase orders. The serial number shall be formulated as follows: three letters abbreviating the manufacturer, followed by two digit number (representing the life raft type), and followed by a 5-digit sequential number starting with 00001 (e.g. EXP2500001 or EXP5000001).

3.9.4 Marking. Marking shall be in English and shall be clearly and legibly applied in a color contrasting to its background, using materials which are permanent for the life of the life raft. Markings shall be located appropriate to the action indicated. Markings shall include:

a. Instructions for manual inflating.

b. Directions for righting if the life raft inflates in an inverted position.

c. Directions for boarding.

d. Position and use of items stowed outside of the equipment container.

e. Contents of survival equipment container(s).

f. “25 PERSONS” or “50 PERSONS”, as applicable, shall be marked over each entrance in characters not less than 100 millimeters in height.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:52Z
Check the source to verify that this is the current version before use.

3.9.5 Code symbols. International Morse Code for the English alphabet A through Z and for Arabic numerals 0 through 9 shall be printed on a 200- by 300-millimeter area or on a piece of canopy fabric and cemented on the inside of the raft canopy adjacent to one entrance. Each code symbol shall be identified by its appropriate letter or numeral counterpart next to the symbol.

3.10 Coated fabric. If a coated fabric different from the fabric used on the USCG-approved commercial 25- or 50-person life rafts is to be used on the proposed life raft design, the coated fabric used in the construction of inflatable buoyancy chambers shall meet the requirements of this section. Polyvinyl chloride (PVC)-coated fabric or PVC alloy-coated fabric shall not be used, regardless of USCG or SOLAS certification. The base fabric shall be inherently rot-proof and shall not be unduly affected by seawater, oil, or fungal attack. Water-soluble materials shall not be used in the coating compound, which shall be applied in a manner such that the finished product consistently meets the performance requirements of this section.

3.10.1 Breaking strength. The breaking strength of the finished coated fabric shall be a minimum of 53 N/mm in warp and fill when tested in accordance with 4.6.1.

3.10.2 Tearing strength. The tearing strength of the finished coated fabric shall be a minimum of 72 newtons in warp and fill when tested in accordance with 4.6.2.

3.10.3 Coating adhesion. The coating adhesion shall be determined by using an air-cure adhesive, as recommended by the fabric manufacturer, to bond one face of the coated fabric to the other face. The minimum adhesion shall be 2.8 N/mm with the coating compound separating off the base fabric when tested in accordance with 4.6.3.

3.10.4 Weld strength of thermoplastic coated fabrics. The coated fabric shall have a minimum peel load of 7 N/mm for the initial break in accordance with 4.6.4.

3.10.5 Resistance to blocking (heat). The coated fabric shall separate in 5 seconds or less when tested in accordance with the procedures of 4.6.5.

3.10.6 Porosity test. The coated fabric shall show no signs of bubble formation indicating leakage when subjected to a maximum pressure of 70 kilopascal gauge (kPag) held for 5.0±0.1 minutes when tested in accordance with 4.6.6.

3.10.7 Coating adhesion after exposure to water. A minimum adhesion result shall be 1.75 N/mm of width when tested in accordance with 4.6.7.

3.10.8 Puncture. The coated fabric shall have a puncture resistance of 360 newtons when tested in accordance with 4.6.8.

3.10.9 Fungus. The coated fabric shall maintain a minimum of 80 percent of original breaking strength and a coating adhesion of 1.75 N/mm when subjected to fungus in accordance with 4.6.9.

3.10.10 Resistance to oil. When sample is tested in accordance with 4.6.10, a single pass of a finger across the sample shall not produce smearing.

3.11 Performance requirements.

3.11.1 Inflation. The Mark-7 and Mark-8 life rafts shall be automatically inflated upon pulling the sea painter. Inflation shall be completed within a period of 30 seconds for the Mark-7 and 45 seconds for the Mark-8 life raft at an ambient temperature of 18 to 20 °C (64 to 68 °F), when tested in accordance with 4.7.3. In addition, inflation time shall be within 30 seconds for the Mark-7 and 45 seconds for the Mark-8 at an ambient temperature of -30 °C (-22 °F), when tested in accordance with 4.7.4. After inflation, the life raft shall maintain its form when loaded with its full complement of persons and equipment.

3.11.2 Over-pressure capability. The inflatable compartment shall be capable of withstanding a pressure equal to at least three times the working pressure and shall be prevented from reaching a pressure exceeding twice the working pressure by means of relief valves when tested in accordance with 4.7.5.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:52Z
Check the source to verify that this is the current version before use.

3.11.3 Environmental exposure. The life raft shall be constructed so as to be capable of withstanding exposure for 30 days afloat in all sea conditions when tested in accordance with 4.7.6.

3.11.4 Operation when dropped. The life raft, in its container, shall be constructed such that, when it is dropped into the water from a height of 23 meters, it shall remain intact and float for a period of 30 minutes. The life raft, when inflated by means of the painter line, shall inflate upright and be boardable when tested in accordance with 4.7.10. The life raft and its equipment shall operate satisfactorily.

3.11.5 Withstanding of jumps. The floating life raft shall be capable of withstanding 25 jumps for the Mark-7 or 50 jumps for the Mark-8 onto it from a height of at least 4.5 meters above its floor. This shall consist of 12 jumps (Mark-7) or 25 jumps (Mark-8) without the canopy erected and 13 jumps (Mark-7) or 25 jumps (Mark-8) with the canopy erected when tested in accordance with 4.7.2. Persons jumping shall weigh not less than 82.5 kilograms.

3.11.6 Towing. The stability of the life raft, when loaded with its full complement of persons and equipment, shall be such that it can be towed at speeds of up to 3 knots in calm water for a distance of 1 kilometer when tested in accordance with 4.7.9.

3.11.7 Service life. The life raft (exclusive of ancillary equipment) and its container shall have an expected service life of at least 20 years when recertified every five years and maintained according to manufacturer's instructions.

3.11.8 Air retention. Air pressure drop shall not exceed 10 percent after compensation for temperature and barometric pressure change when tested in accordance with 4.7.1.

3.11.9 Loading and seating. The loading and seating shall be determined using the procedure in 4.7.7 to verify the Mark-7 life raft can accommodate 25 persons and the Mark-8 life raft can accommodate 50 persons.

3.11.10 Canopy closure. There shall be not more than 20 liters of accumulated water when tested in accordance with 4.7.8.

3.11.11 Function/operation. The life raft shall inflate at the required pull force as specified in 3.6.1.b. During inflation, the life raft shall be observed for impediment or blockage of gas flow, loss of gas to the ambient air, or restriction by any component or accessory. There shall be no evidence of structural or material failure in any respect. All sealed areas, seams, seam tapes, and attachments shall remain intact and shall show no indication of separation (i.e. free of voids, lifting seams, etc.).

3.12 Servicing.

3.12.1 Frequency. The life raft shall be designed to be serviced once every five years at approved Navy servicing facilities.

3.12.2 Service manual. A service manual covering the opening, inspection, testing, repair, and repacking of the approved life raft with parts list of all replaceable/repairable components shall be provided, as specified (see 6.2).

## 4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on one life raft, complete with equipment and stowage container, when a first article sample is required (see 3.1). This inspection shall include the examination of 4.4 and the tests of 4.7.1 through 4.7.10, as indicated in table II.

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.4, 4.5, and the tests of 4.7.1 through 4.7.10, as indicated in tables II through V. Reliability based on sampling shall be 80 percent with a minimum confidence level of 0.5 in accordance with the Cumulative Binomial Distribution Nomograph.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:52Z
Check the source to verify that this is the current version before use.

TABLE II. First article tests and conformance inspections.

| Characteristic | Requirement Paragraph | Test Paragraph | First Article Test | Conformance Inspection | Verification Method |
|---|---|---|---|---|---|
| Shape | 3.4.1 | 4.4 | X | -- | Visual Inspection |
| Stability | 3.4.3 | 4.4 | X | -- | Visual Inspection |
| Buoyancy Chambers | 3.4.4.1 | 4.4 | X | -- | Visual Inspection |
| Storage Pockets | 3.4.4.2 | 4.4 | X | -- | Visual Inspection |
| Reinforcing Patches | 3.4.4.3 | 4.4 | X | X | Visual Inspection |
| Inflation Cylinder Holders | 3.4.4.4 | 4.4 | X | -- | Visual Inspection |
| Exposed Surfaces | 3.4.4.5 | 4.4 | X | X | Visual Inspection |
| Floor | 3.4.5 | 4.4 | X | X | Visual Inspection |
| Canopy | 3.4.6 | 4.4 | X | X | Visual Inspection |
| Entrances | 3.4.7 | 4.4 | X | X | Visual Inspection |
| Laps and Seams | 3.4.8 | 4.6.3 and 4.6.4 | X | X | Visual Inspection & Test Documentation |
| Retro-Reflective Material | 3.4.10 | 4.4 | X | X | Visual Inspection |
| Color | 3.4.11 | 4.4 | X | X | Visual Inspection |
| Container | 3.4.12 | 4.4 | X | X | Visual Inspection |
| Inflation | 3.4.13 | 4.4 | X | X | Demonstration |
| Boarding Ramps | 3.5.1 | 4.4 | X | X | Visual Inspection |
| Boarding Ladder | 3.5.2 | 4.4 | X | X | Visual Inspection |
| Exterior Canopy Lamp | 3.5.3 | 4.7.3 | X | -- | Demonstration & Test Documentation |
| Interior Canopy Lamp | 3.5.4 | 4.7.3 | X | -- | Demonstration & Test Documentation |
| Lifelines | 3.5.5 | 4.4 | X | X | Visual Inspection & Test Documentation |
| Sea Painter System | 3.5.6 | 4.4 | X | X | Visual Inspection |
| Righting Gear | 3.5.7 | 4.4 | X | X | Visual Inspection |
| Towing/Mooring Connections | 3.5.8 | 4.4 | X | X | Visual Inspection & Test Documentation |



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:52Z
Check the source to verify that this is the current version before use.

TABLE II. First article tests and conformance inspections – Continued.

| Characteristic | Requirement Paragraph | Test Paragraph | First Article Test | Conformance Inspection | Verification Method |
|---|---|---|---|---|---|
| Inflation Cylinders | 3.6.1 | 4.4 | X | X | Visual Inspection & Test Documentation |
| Inflation Cylinders | 3.6.1.a | 4.4 | X | -- | Test Documentation |
| Inflation Cylinders | 3.6.1.b | 4.7.3 | X | X | Demonstration |
| Inflation Valve | 3.6.2 | 4.4 | X | | Test Documentation |
| Pressure Relief Valve Poppet Retainer | 3.6.3 | 4.7.5 | X | -- | Demonstration |
| Valves | 3.6.4 | 4.4 | X | X | Visual Inspection & Test Documentation |
| Buoyant Rescue Quoit and Heaving Line | 3.7.1 | 4.4 | X | X | Visual Inspection |
| Knife | 3.7.2 | 4.4 | X | X | Visual Inspection |
| Bailers | 3.7.3 | 4.4 | X | X | Visual Inspection |
| Cellulose Spongers | 3.7.4 | 4.4 | X | X | Visual Inspection |
| Sea Anchor | 3.7.5 | 4.4 | X | X | Visual Inspection |
| Buoyant Paddles | 3.7.6 | 4.4 | X | X | Visual Inspection |
| Pump or Bellows | 3.7.7 | 4.4 | X | X | Visual Inspection |
| FAK | 3.7.8 | 4.4 | X | X | Visual Inspection |
| Abandon Ship Signal Kit | 3.7.9 | 4.4 | X | X | Visual Inspection |
| Electric Flashlight | 3.7.10 | 4.4 | X | X | Visual Inspection & Test Documentation |
| Mirror, Signaling | 3.7.11 | 4.4 | X | X | Visual Inspection |
| Whistle | 3.7.12 | 4.4 | X | X | Visual Inspection |
| Lifesaving Signal Waterproof Card | 3.7.13 | 4.4 | X | X | Visual Inspection |
| Fishing Tackle | 3.7.14 | 4.4 | X | X | Visual Inspection |
| Food Rations | 3.7.15 | 4.4 | X | X | Visual Inspection & USCG Certification |
| Drinking Water | 3.7.16 | 4.4 | X | X | Visual Inspection & USCG Certification |
| MROD | 3.7.17 | 4.4 | X | X | Visual Inspection |
| Anti-Seasickness Medicine | 3.7.18 | 4.4 | X | X | Visual Inspection |
| Survival Instructions | 3.7.19 | 4.4 | X | X | Visual Inspection |



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:52Z
Check the source to verify that this is the current version before use.

TABLE II. First article tests and conformance inspections – Continued.

| Characteristic | Requirement Paragraph | Test Paragraph | First Article Test | Conformance Inspection | Verification Method |
|---|---|---|---|---|---|
| Immediate Action Instructions | 3.7.20 | 4.4 | X | X | Visual Inspection |
| Thermal Protective Aid | 3.7.21 | 4.4 | X | X | Visual Inspection & Test Documentation |
| Repair Outfit | 3.7.22 | 4.4 | X | X | Visual Inspection |
| Spare Battery Pack | 3.7.23 | 4.4 | X | X | Visual Inspection |
| Towing/Mooring Connection Hooks | 3.7.24 | 4.4 | X | X | Visual Inspection |
| Survival Equipment Bag(s) | 3.7.25 | 4.4 | X | X | Visual Inspection |
| Size and Mass | 3.8 | 4.4 | X | | Test Documentation |
| Life Raft Nameplate | 3.9.1 | 4.4 | X | X | Visual Inspection |
| Life Raft Container Data Plate | 3.9.2 | 4.4 | X | X | Visual Inspection |
| Serial Numbers | 3.9.3 | 4.4 | X | X | Visual Inspection |
| Marking | 3.9.4 | 4.4 | X | X | Visual Inspection |
| Code Symbols | 3.9.5 | 4.4 | X | X | Visual Inspection |
| Coated Fabric (if applicable) | 3.10.1 – 3.10.10 | 4.5 and 4.6 (if applicable) | X | -- | Test Documentation |
| Inflation | 3.11.1 | 4.7.3 | X | -- | Demonstration & Test Documentation |
| Over-Pressure Capability | 3.11.2 | 4.7.5 | X | -- | Test Documentation |
| Environmental Exposure | 3.11.3 | 4.7.2 | X | -- | Test Documentation |
| Operation When Dropped | 3.11.4 | 4.7.10 | X | -- | Test Documentation |
| Withstanding of Jumps | 3.11.5 | 4.7.2 | X | -- | Test Documentation |
| Towing | 3.11.6 | 4.7.9 | X | -- | Test Documentation |
| Service Manual | 3.12.2 | 4.4 | X | -- | Visual Inspection |
| Typical Life Raft Defects | Table III | 4.4 | | X | Visual Inspection |
| Air Retention | 3.11.8 | 4.7.1 | X | X | Test Documentation |
| Loading and Seating | 3.11.9 | 4.7.7 | X | -- | Test Documentation |
| Canopy Closure | 3.11.10 | 4.7.8 | X | -- | Test Documentation |
| Function/Operation | 3.11.11 | 4.4 | X | X | Visual Inspection |

4.4 Examination. Each life raft shall be examined for compliance with the requirements specified in 3.2 through 3.10 using table III. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:52Z
Check the source to verify that this is the current version before use.

TABLE III. Types of typical life raft defects.

| TYPICAL LIFE RAFT DEFECTS |
|---|
| 1. Hoses damaged. End connections not torqued as specified in Service Manual. End connections cross-threaded. |
| 2. Evidence of sewing in the seams of, or through, the fabric of air chambers as specified in 3.4.8. |
| 3. Valve pull-cables damaged or not connected as specified in Service Manual. |
| 4. Topping-off valves not installed correctly as specified in 3.6.4. |
| 5. Inflation cylinders not fully charged to the weight and pressure specified in Service Manual. |
| 6. Banding straps not installed as specified in 3.4.12. |
| 7. Painter rope missing, not attached, not as specified in Service Manual. |
| 8. Pull force on painter line to activate the inflation system exceeds 250 Newtons as specified in 3.6.1.b. |
| 9. Hand pump or bellows and adapters not operable as specified in 3.7.7. |
| 10. Boarding ladder damaged, not as specified in 3.5.2. |
| 11. Canopy damaged, not as specified in 3.4.6. |
| 12. Sea anchors missing, not installed or stowed as specified in 3.7.5. |
| 13. Cylinder holders not installed as specified in 3.4.4.4. |
| 14. Weight of the completed, packed life raft, ready for stowage exceeds maximum weight specified in 3.8. |
| 15. Materials used other than those specified. Survival gear missing or not as specified in 3.7. |
| 16. Equipment incomplete, components not as specified, not stowed as specified in 3.7. |
| 17. Length and width of the finished life raft not within the tolerances specified in Service Manual. |
| 18. Hull, gunwale, cross tube, diameters not within the tolerances specified in Service Manual. |
| 19. Unauthorized repairs or rework (see 3.4.14). |
| 20. Paddles damaged or missing, not as specified in 3.7.6. |
| 21. Lights not installed as specified in 3.5.3 and 3.5.4, batteries missing. |
| 22. Canopy closure fasteners not installed, not as specified in 3.4.6. |
| 23. Righting gear missing, not installed as specified, not as specified in 3.5.7. |
| 24. Floatable knives missing, not as specified in 3.7.2. |
| 25. Inner canopy vent holes not as specified in 3.4.6. |
| 26. Identification marking not legible, not as specified in 3.9.1. |
| 27. Instruction cards missing, not as specified in 3.7.19. |
| 28. Caution tag or instruction lettering missing, not as specified in 3.7.20. |
| 29. Rope or line ends not fused or whipped as specified in 3.4.14. |
| 30. Rain catcher as specified in 3.4.6 missing. |
| 31. Inflatable/insulated floor not as specified in 3.4.5. |
| 32. Sealing clamps missing or not as specified in 3.7.22. |
| 33. Buoyant rescue quoit and heaving and line not as specified, not installed as specified in 3.7.1. |
| 34. Lifeline not as specified, not installed as specified in 3.5.5. |



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:52Z
Check the source to verify that this is the current version before use.

4.5 Material inspection.

a. Examination of the coated and cured fabric. Defects found during the examination shall be classified in accordance with item e below and table IV.

b. Meter-by-meter examination. The required meterage of each roll shall be inspected and visual defects classified as listed in item e below. The material shall be examined on one side; however, the side examined shall be alternated from roll to roll. The material shall be through-lighted and examined for pin holes. The defects shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the fabric, in which case only the more serious defects shall be counted. A continuous defect shall be counted as one defect for each warp-wise meter or fraction thereof in which it occurs. The unit of product for this examination shall be 1 linear meter. The number of pieces from which the sample is to be selected shall be in accordance with table V. An approximate equal number of meters shall be examined in each piece of the sample.

c. Meter-by-meter fabric examination defects. A list of defects that may be present are listed in table IV.

d. Overall examination. Each defect listed in table IV shall be counted not more than once in each piece examined. The unit of product for this examination shall be one piece. The sample size shall be in accordance with table V. The lot shall be unacceptable if one or more of the defects listed are found:

e. Overall examination defects.

(1) Wrong color.

(2) Not clean throughout.

(3) Adhering to successive layer on roll (resists unrolling), tacky, cracked, or brittle.

(4) Objectionable odor.

(5) Variation in weight. The variation in mass shall be not greater than ±50 grams per meter squared of the approved coated fabric.

TABLE IV. Meter-by-meter fabric examination defects.

| **EXAMINATION DEFECTS** | |
|---|---|
| Abrasion mark | Resulting in a weak place or evidence of coating removed (clearly visible at normal inspection distance [about 1 meter]) |
| Blister lump or tunnel | Any soft (hollow), clearly noticeable |
| Crease or wrinkle embedded | Or resulting in doubling, or adhesion of surfaces against each other, or incomplete coating |
| Cut (including cuts through coating only), hole, tear, or pinhole | Any light areas or windows (poor distribution of pigments) |
| Uneven coating | Thin areas where coating compound is missing or insufficient; or heavily coated areas, noticeably stiffer than adjoining, unaffected fabric |
| Selvedge | Scalloped, deepest indentation 15 millimeters or more: tight, resulting in pucker or waviness in adjoining fabric |
| Uneven tension | Puckers, waviness, ridges that will not flatten under manual pressure |
| Spot, stain, or streak | Other than coating or pigment and more than 15 millimeters in its longest dimension |
| Width | Less than specified |



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:52Z
Check the source to verify that this is the current version before use.

TABLE V. Sampling for examination.

| LOT SIZE (METERS) | SAMPLE SIZE (IN PIECES) |
|---|---|
| Up to 1,300 | 3 |
| 1,301 to 3,200 | 5 |
| 3,201 to 8,000 | 8 |
| 8,001 to 22,000 | 10 |
| 22,001 and up | 15 |
| NOTE: If the lot contains fewer than three pieces, each piece of that lot shall be examined. | |

4.6 Coated fabric. The following tests shall be conducted when a coated fabric is used that is different from the fabric used on the USCG-approved commercial 25- or 50-person life rafts, as specified in 3.10.

4.6.1 Breaking strength. The finished coated fabric shall be tested in accordance with ASTM D5035, specimen type 1C-E. The breaking strength of the finished coated fabric shall be a minimum of 53 N/mm in warp and fill.

4.6.2 Tearing strength. The finished coated fabric shall be tested in accordance with ASTM D2261. The tearing strength of the finished coated fabric shall be a minimum of 72 Newtons in warp and fill.

4.6.3 Coating adhesion. Life rafts fabricated from thermoplastic materials and using RF welding technology for seams and attachments shall undergo the coating adhesion test. The coating adhesion shall be determined by using an air-cure adhesive that is compatible with thermoplastic materials to bond one face of the coated fabric to the other face. The minimum adhesion shall be 2.8 N/mm with the coating compound separating off the base fabric. Use testing parameters in accordance with ASTM D751 (see note 17 for coated fabrics).

4.6.4 Weld strength of thermoplastic coated fabrics. Life rafts fabricated from thermoplastic materials and using RF welding technology for seams and attachments shall undergo a weld strength test in accordance with ASTM D751 (see note 17 for coated fabrics). Ensure the following:

a. There shall be two samples of fabric (300 by 200 millimeters each), cut with a short side parallel to the warp direction and superimposed face to back for double-coated fabrics, or coated face to coated face for single-coated fabrics.

b. The fabric samples shall be high frequency (HF) welded with a tool to a 10±1-millimeter width and convenient length.

c. Twenty-five-millimeter-wide test specimens shall be cut transversely to the line of weld.

d. The maximum peel load shall be recorded. The minimum peel load shall be 7 N/mm for the initial break.

4.6.5 Resistance to blocking (heat). The coated fabric shall separate in 5 seconds or less when tested in accordance with the following procedure:

a. Place two 150- by 25.4-millimeter coated fabric specimens in an oven on a smooth surface in such a manner that the ends are overlapped 25.4 millimeters.

b. Place a 2.722-kilogram weight directly on the overlapped area.

c. After conditioning at a temperature of 70±2 °C for 4 hours, remove the weight and take the specimens from the oven and condition for 1 hour at 22.7±3 °C.

d. Attach one end of the specimen in a clamping device, allowing the free end to hang down.

e. Suspend a 113.4-gram weight from the free end of the specimen.

f. Record time required to separate.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:52Z
Check the source to verify that this is the current version before use.

4.6.6 Porosity test. Unless otherwise specified (see 6.2), five specimens from each inspection lot shall be tested. The coated fabric shall not have any bubbles in 5 minutes when tested as follows:

a. Porosity shall be determined by a test jig as shown on figure 2.

b. The test specimen shall have a diameter of not less than 325 millimeters.

c. The specimen shall be placed, outer side up, on the holder and the plate collar tightly bolted thereon. Care shall be taken to ensure that there is no leakage at the seal.

d. A sufficient quantity of water shall be poured on the specimen to keep it completely covered under maximum pressure.

e. Pressure shall be increased to 70 kPag.

f. Air bubbles on the surface of the specimen produced by air pressure closing the spaces between the fabric and the coating shall be removed.

g. The pressure shall be held at 70 kPag for 5.0±0.1 minutes.

h. At the end of the 5-minute period, the specimen shall be examined for evidence of bubble formation, indicating leakage.

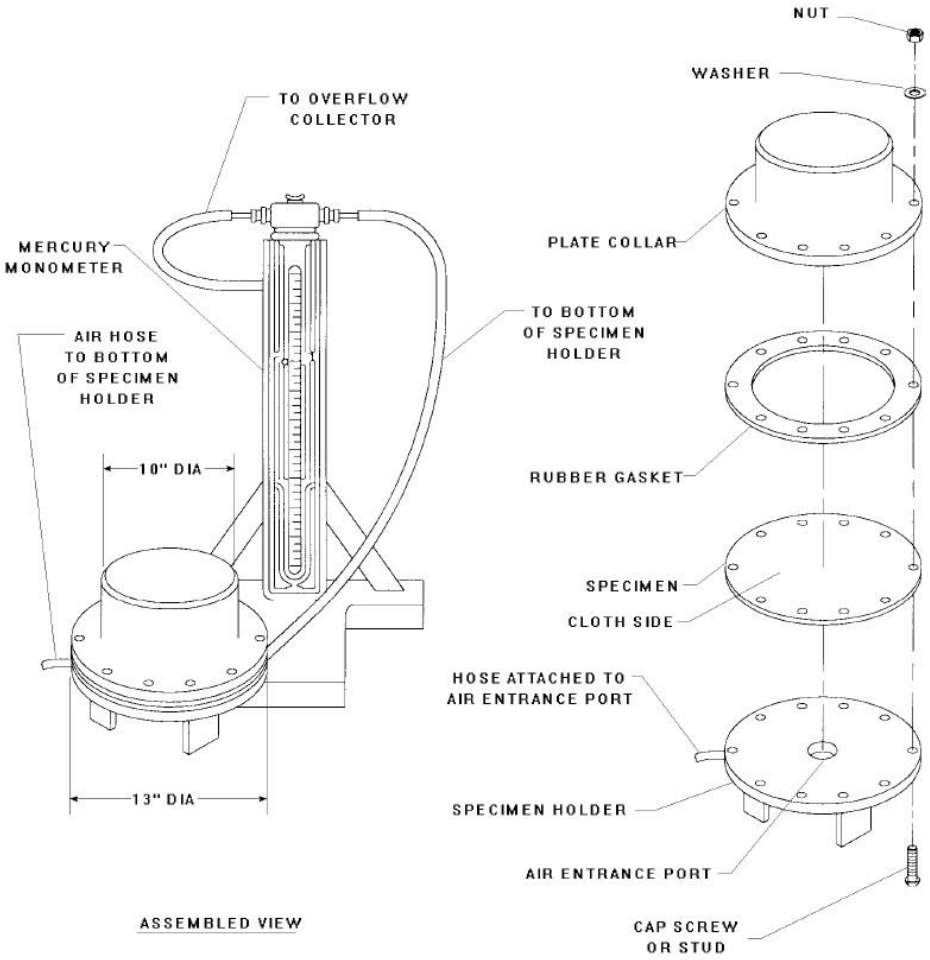



FIGURE 2. Air leakage inspection apparatus.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:52Z
Check the source to verify that this is the current version before use.

4.6.7 Coating adhesion after exposure to water.

a. Three test samples shall be made in accordance with ASTM D751 (see note 17 for coated fabrics), except the adhesive system used for the samples shall be the same system used for production parts.

b. Test specimens for coating adhesion shall be immersed in 3 percent sodium chloride water solution at 60±1 °C for 96±1 hour.

c. The edges of the test specimens shall not be sealed.

d. The test specimens shall be removed from the warm water and immediately placed in water at room temperature.

e. Each determination shall be completed within 3 minutes of the time of removal from the water. Minimum adhesion results shall be 1.75 N/mm of width.

4.6.8 Puncture. The coated fabric shall have a puncture resistance of 360 Newtons when tested as follows:

a. The test shall be conducted in accordance with ASTM D3787, except that the ring clamp mechanism shall have an internal diameter of 76 millimeters, and the ball shall be replaced by a piercing instrument shaped like a flared, flat-tip screwdriver, having a width of 7.93±0.250 millimeters and a thickness of 0.79±0.10 millimeters at the extreme tip.

b. The piercing tip edges shall be rounded to a 0.25-millimeter radius.

c. The piercing instrument shall be oriented to intercept the warp and fill threads at an angle of approximately 45 degrees.

d. The average of three test specimens shall be reported.

4.6.9 Fungus. The coated fabric shall be tested in accordance with Method 508-8 of MIL-STD-810, except that the test specimen shall not be sterilized before being subjected to the fungus. The coated fabric shall maintain a minimum of 80 percent of original breaking strength and a coating adhesion of 1.75 N/mm.

4.6.10 Resistance to oil. The coated fabric shall be tested in accordance with the following method:

a. The test shall be carried out not less than 16 hours after curing.

b. Each coated face shall be tested for 22 hours at 20±1 °C.

c. The test specimen shall be folded over so that the surfaces are pressed together and there are no signs of residual tackiness.

d. With the test specimen opened out, a single pass of the finger over the exposed surface shall not produce smearing.

e. Disk specimen for testing shall be at least 70 millimeters in diameter. A typical apparatus required is shown in figure 3. It consists of a base plate (A) and an open-ended cylindrical chamber (B), which is held tightly against the test specimen (C) by wing nuts (D). The wing nuts are mounted on bolts (E). A hole with a diameter of approximately 30 millimeters shall be made in the base plate for the examination of the surface not in contact with the liquid. During the test, the opening in the top of the chamber is closed by a close-fitting plug (P).

f. Test specimens in the “As Received” condition shall be conditioned for not less than 3 hours at 20±2 °C immediately before testing.

g. The test specimens shall be placed in the apparatus as indicated on figure 3. The chamber of the apparatus is then filled with the test liquid to a depth of approximately 20 millimeters and the plug (P) inserted.

h. At the end of the contact period, the test liquid is removed and the test specimen is released. Any surplus liquid is removed from the surface of the test specimen by blotting with filter paper or a textile fabric that does not deposit lint.

i. The test specimen shall be folded over so that the surfaces are pressed together to see if there are any signs of residual tackiness.

j. With the test specimen opened out, a single pass of the finger over the exposed surface shall not produce smearing.

k. The liquid to be used as a standard test oil shall be IRM 901 Oil in accordance with ASTM D471.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:52Z
Check the source to verify that this is the current version before use.

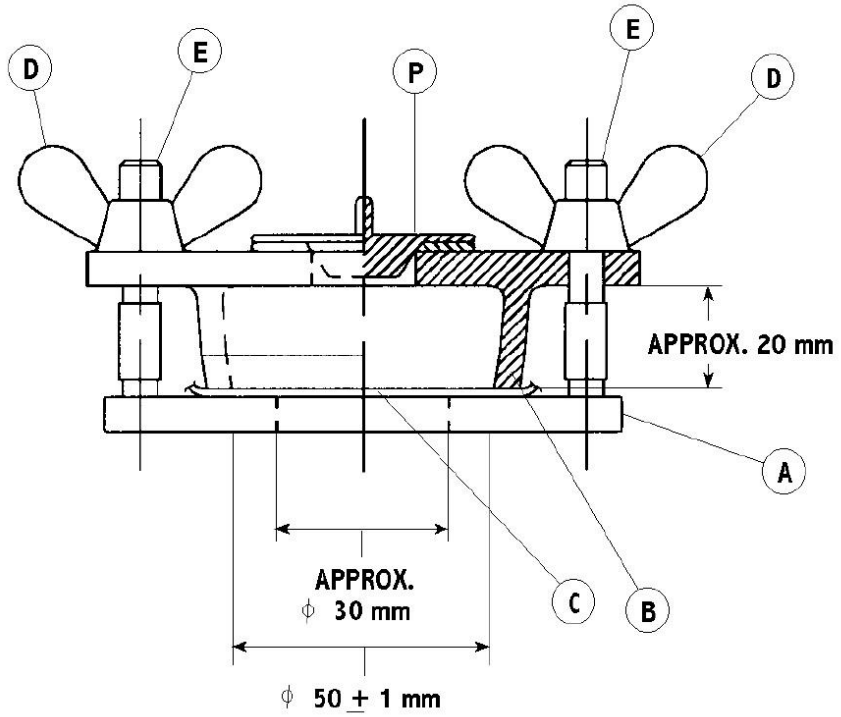



FIGURE 3. Apparatus for test with liquid on one surface only.

4.6.11 Hydrolysis resistance. When tested by the following test methods, performance values shall be achieved:

a. Coating adhesion: 70 percent of original value.

b. Blocking resistance: Separation in 5 seconds when tested as follows:

(1) Place two coated fabric specimens, each 150 by 25.4 millimeters, in an oven on a smooth surface in such a manner that the ends are overlapped 25.4 millimeters.

(2) Place a 2.722-kilogram weight directly on the overlapped area.

(3) After conditioning at a temperature of 70±2 °C for 4 hours, remove the weight and take the specimens from the oven and condition for 1 hour at 22.7±3 °C.

(4) Attach one end of the specimen in a clamping device, allowing the free end to hang down.

(5) Suspend a 113.4-gram weight from the free end of the specimen. Record the time required to separate.

c. Weld strength: 70 percent of recorded values when tested as follows:

(1) Two samples of fabric (300 by 200 millimeters), cut with a short side parallel to the warp direction, shall be superimposed face to back for double-coated fabrics or coated face to coated face for single-coated fabrics.

(2) The fabric samples shall be HF-welded with a tool to a 10±1-millimeter width and convenient length.

(3) Twenty-five-millimeter-wide test specimens shall be cut transversely to the line of weld.

(4) The test samples shall be in accordance with ASTM D751 (see note 17 for coated fabrics).

(5) The maximum peel load shall be recorded.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:52Z
Check the source to verify that this is the current version before use.

4.6.12 Resistance to accelerated weathering. The exterior side of the coated fabric and representative seam samples shall be exposed in an Atlas Weatherometer or equivalent in accordance with AATCC TM169, Option 4.

a. Coating shall not crack when the material is bent, more heavily coated side out, over a 3-millimeter diameter rod, after exposure for 100 hours without "Corex D" filters or equivalent. Only one side of the fabric shall be tested, which shall be the side facing the arc in the testing apparatus. After weathering, the coated fabric and representative seam samples shall show a loss in initial tensile strength not exceeding 10 percent when tested in accordance with ASTM D5035, specimen type 1C-E.

b. The resistance to blocking (heat) shall be tested as follows:

(1) Place two coated fabric specimens, each 150 by 25.4 millimeters, in an oven, on a smooth surface, in such a manner that the ends are overlapped 25.4 millimeters.

(2) Place a 2.722-kilogram weight directly on the overlapped area.

(3) After conditioning at a temperature of 70±2 °C for 4 hours, remove the weight and take the specimens from the oven and condition for 1 hour at 22.7±3 °C.

(4) Attach one end of the specimen in a clamping device, allowing the free end to hang down.

(5) Suspend a 113.4-gram weight from the free end of the specimen.

(6) Record the time required to separate.

c. Single-coated fabrics shall be subjected to a coating adhesion test (see 4.6.3) after the weatherometer test and shall have a minimum of 70 percent of the coating adhesion.

d. Upon completion of steps a and b above, the test sample shall be folded and shall exhibit no cracks, delamination, or visual deterioration when samples are inspected under a magnification of 2×.

4.6.13 Abrasion resistance. The exterior side of the inflatable fabric shall withstand a minimum of 25,000 cycles of chafing without exposure of the fabric substrate when the fabric is coated on one side only. When the fabric is coated on both sides and the inside coating is at least equal to or greater than the exterior coating, the exterior side shall withstand a minimum of 15,000 cycles without exposure of fabric substrate. Testing shall be performed in accordance with ASTM D3884.

4.6.13.1 Coated fabric with USCG/SOLAS approval. If a coated fabric that has already received USCG/SOLAS approval is used, the test requirements of 4.6.13 will not be required to be met.

4.6.13.2 Coated fabric without USCG/SOLAS approval. If a coated fabric that has not received USCG/SOLAS approval is used, the minimum requirements of 4.6.13 shall be met.

4.6.14 Seam tests. The bonding together of any two or more pieces of coated fabric (such as lap joints, butt joints, and closures; chafing or reinforcing patches; and coated fabric flanges of valves) shall be considered a seam. Seams shall be subjected to all seam tests specified in table II (except for chafing patches, which will only be tested for peel) as follows:

a. The average breaking strength of five specimens for each type of seam for each test shall be compared to table II for conformance.

b. Breaking strength specimens shall be 50 millimeters wide (parallel to the seam) and shall extend (perpendicular to the seam) 75 millimeters beyond both edges of the seam.

c. For tests conducted after immersion in a fluid, no part of the test specimens shall be coated or covered during the water immersion periods.

d. Specimens shall be cooled in the immersion fluids at 23±3 °C for 30 to 90 minutes before testing.

e. Testing of immersed specimens shall be completed within 3 minutes after removal from the immersion fluids.

f. The average peel adhesion strength of two specimens for each type seam shall be compared to table II for conformance.

g. Peel adhesion specimens shall be of sufficient length to determine both the initial and after-water adhesion values on the same specimen.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:52Z
Check the source to verify that this is the current version before use.

4.6.15 Dead load shear resistance. The dead load shear resistance of the seams shall be tested in accordance with the following procedure:

a. Three specimens shall be tested for each seam type.

b. The test specimens shall be 25.4±0.050 millimeters wide (parallel to the seam) and coated fabric shall extend a minimum of 75 millimeters (perpendicular to the seam) on each side of the seam.

c. One index mark shall be scribed on each side of the seam to facilitate observation and measurement of slippage.

d. Each specimen shall be subjected to a constant (dead load) force of 23±0.25 kilograms at 93±3 °C.

e. After 24 hours, each specimen shall be examined while still under load for signs of slippage or separation as stated in table VI.

TABLE VI. Characteristics of seams.

| Property | Requirements | Test Method |
|---|---|---|
| Breaking Strength (initial) | 53 N/mm (min) | ASTM D751 [1/] and as specified herein |
| Breaking Strength – after immersion in distilled water at 71±1 °C for the following duration:<br>- 14 days<br>- 42 days | <br><br>49 N/mm<br>25 N/mm | ASTM D751, ASTM D471, and as specified herein |
| Dead Load Shear Resistance – under 23 kg/mm in stress at 93.3 °C (200 °F) for 24 hours | 2.5 mm slippage (max) | As specified herein |
| Peel Adhesion (initial) | 3.5 N/mm (min) | ASTM D413 machine method and as specified herein |
| Peel Adhesion – after immersion in distilled water at 71±1 °C for the following duration.<br>- 14 days<br>- 42 days | <br><br>1.75N/mm<br>1.40 N/mm | ASTM D413 machine method, ASTM D471, and as specified herein |
| NOTE:<br>[1/] Except that the specimen shall be 50.8 millimeters in width. | | |

4.7 Life raft tests.

4.7.1 Air retention test. Each life raft shall pass an air retention test under the following conditions:

a. The testing period shall be 4 hours, with a maximum allowable pressure drop of 10 percent after compensation for ambient temperature and barometric pressure changes.

b. For each degree Celsius rise in temperature, 0.385 kilopascal shall be subtracted from the final pressure reading. For each degree Celsius drop in temperature, 0.385 kilopascal shall be added to the final pressure reading.

c. For each millimeter of mercury rise in barometric pressure during the 4-hour period, 0.133 kilopascal shall be added to the final temperature-corrected pressure reading. For each millimeter of mercury drop in barometric pressure, 0.133 kilopascal shall be subtracted from the final temperature-corrected pressure reading.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:52Z
Check the source to verify that this is the current version before use.

d. One inflatable life raft from each lot of less than 30 life rafts and two from each lot of 30 or more life rafts shall pass the requirements of 4.7.1. If any life raft fails this test:

(1) The reason for the failure shall be determined.

(2) Each life raft in the lot shall be examined for the defect.

(3) The lot test shall be repeated, including random selection of the life raft or life rafts to be tested.

4.7.2 Jump test. Thirteen (Mark-7) or 25 (Mark-8) of the jumps shall be with the canopy erect and 12 (Mark-7) or 25 (Mark-8) of the jumps with the canopy furled or deflated. The test subject shall not weigh less than 82.5 kilograms. If an equivalent mass is used, it shall be equipped with the shoes described in Part 1, Regulation 5.2.1 of IMO Resolution MSC.81(70) arranged to strike the life raft first.

4.7.3 Inflation actuation test – ambient temperature. The Mark-7 and Mark-8 life rafts shall be tested to validate the ability of the life rafts to inflate by sea painter actuation of the inflation system. This test shall be performed at a temperature range of 18 to 20 °C using the following procedure:

a. The rubber plug shall be removed from the sea painter bung assembly.

b. The sea painter shall be pulled from the sea painter bung assembly until resistance is felt.

c. A bight shall be tied on to the sea painter a safe distance from the life raft, so test personnel are not struck by the inflating life raft.

d. A calibrated full force gauge set to peak hold shall be attached to the bight in the sea painter.

e. Time shall be started on a stop watch, at which time the test person shall exert pull on the pull force gauge attached to the bight in the sea painter. The pull shall be steady and kept parallel with the floor and centerline of the life raft container until the inflation system is actuated.

f. The inflation of the life raft shall be observed to ensure the life raft fully inflates with arch and canopy up, all boarding ramps inflated, pressure relief valves venting, exterior canopy lamp(s) flashing, and interior canopy lamp(s) illuminated. When the life raft is fully inflated, timing with stop watch shall halt.

g. After the pressure relieve valves have stopped venting and the temperature of the life raft has been stabilized, measure the pressure in the hull chambers.

h. A successful inflation test results in complete inflation in less than 30 seconds for the Mark-7 and 45 seconds for the Mark-8 life raft (see 3.11.1), with an actuation pull force of 250 Newtons or less (see 3.6.1.b), hull chambers air pressure attaining the life raft working pressure (see 3.6.4); and exterior canopy lamp(s) (see 3.5.3) and interior canopy lamps(s) (see 3.5.4) illuminated.

4.7.4 Inflation actuation test – cold temperature extreme. The Mark-7 and Mark-8 life rafts shall be tested to validate the ability of the life rafts to inflate by sea painter actuation of the inflation system. This test shall be performed at a temperature of -30 °C using the following steps:

a. The life raft shall be cold soaked to a temperature of -30 °C for a period not less than 24 hours.

b. The life raft shall be removed from the environmental chamber.

c. The rubber plug shall be removed from the sea painter bung assembly.

d. The sea painter shall be pulled from the sea painter bung assembly until resistance is felt.

e. A bight shall be tied on to the sea painter a safe distance from the life raft so test personnel are not struck by the inflating life raft.

f. A calibrated full force gauge set to peak hold shall be attached to the bight in the sea painter.

g. Time shall be started on a stop watch, at which time the test person shall exert pull on the pull force gauge attached to the bight in the sea painter. The pull shall be steady and kept parallel with the floor and centerline of the life raft container until the inflation system is actuated.

h. The inflation of the life raft shall be observed to ensure the life raft fully inflates with arch and canopy up, all boarding ramps inflated, pressure relief valves venting, exterior canopy lamp(s) flashing, and interior canopy lamp(s) illuminated.

i. When the life raft is fully inflated, timing with stop watch shall halt.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:52Z
Check the source to verify that this is the current version before use.

j. After the pressure relieve valves have stopped venting and the temperature of the life raft stabilized, measure the pressure in the hull chambers.

k. A successful inflation test results in complete inflation in less than 30 seconds for the Mark-7 and 45 seconds for the Mark-8 life raft (see 3.11.1) with an actuation pull force of 250 Newtons or less (see 3.6.1.b), hull chamber air pressure attaining the life raft working pressure (see 3.6.4); and exterior canopy lights (see 3.5.3) and interior canopy lights (see 3.5.4) illuminated.

4.7.5 Over-pressure test. The hull and arch chambers shall be subjected to a 300 percent working pressure test (see 3.11.2) conducted as follows:

a. All pressure relief valves shall be disabled with a poppet retainer (see 3.6.3). The pressure of each hull and arch chamber shall be increased to 300 percent the working pressure of the life raft and held for 5 minutes. During that time, no failure of the air retention integrity of the chambers shall occur.

b. The poppet retainers shall then be removed to allow the chamber to vent the additional pressure.

c. No failure of the seams, fabric, or valves shall occur during the test.

4.7.6 Mooring-out test. The life raft shall be loaded with a mass equal to the total number of personnel for which the life raft is rated (Mark-7: 2,063 kilograms, Mark-8: 4,125 kilograms) and its equipment. The life raft shall be moored at sea or in a seawater harbor. The life raft may be topped-off once per day with the manual inflation pump; however, it should retain its shape over any 24-hour period. Initial inflation shall be compressed air. The life raft shall remain afloat for 30 days without any damage or degradation that would impair its performance.

4.7.7 Loading and seating test. The life raft shall be tested for loading and seating in accordance with Part 1, Regulation 5.7 of IMO Resolution MSC.81(70). The persons used to determine seating capacity shall wear insulated buoyant immersion suits rather than lifejackets.

4.7.8 Canopy closure test. Canopy closure shall be tested in accordance with Part 1, Regulation 5.12 of IMO Resolution MSC.81(70). The water accumulated within the life raft at the end of the testing shall not exceed 20 liters.

4.7.9 Towing test. The life raft, with full complement of persons and equipment, shall be able to be towed at a speed of 3 knots for 1 kilometer when subjected to a towing test in accordance with Part 1, Regulation 5.4 of IMO Resolution MSC.81(70).

4.7.10 Drop test. The life raft, container, and survival equipment shall survive a drop of 23 meters from the water surface with no damage to the container, life raft, and survival equipment and remain afloat for 30 minutes when subjected to a drop test in accordance with Part 1, Regulation 5.1 of IMO Resolution MSC.81(70), except that the drop height shall be 23 meters from the water's surface.

## 5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:52Z
Check the source to verify that this is the current version before use.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The life rafts described herein are intended for shipboard installation to be used in the event the ship is abandoned at sea. These life rafts are military unique since they have a container, sea painter, flares, survival manual, inflation system, and a Government-furnished manual reverse osmosis desalinator. Additionally, they have a military specific drop height and certification periodicity.

6.1.1. Use of new technology. Equipment manufactured in accordance with this document may utilize new technology provided form, fit, and function are maintained as approved by the Technical Authority.

6.1.2. Current stock of FAK. Current stock of FAK may be used until depleted; future requisitions for FAK should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When first article inspection is required (see 3.1 and 6.10).
c. Requirements for repair or rework (see 3.4.14 and 6.11).
d. Government-furnished property (GFP) (see 3.7.9, 3.7.17, 3.7.19, and 6.3).
e. Training material and maintenance instruction requirements (see 3.12).
f. Service Manual requirements (see 3.12.2).
g. When other than 5 specimens are tested as part of the porosity test (see 4.6.6).
h. Packaging requirements and additional packaging considerations (see 5.1 and 6.9).
i. Shelf-life requirements (see 6.5).
j. Responsibility for inspection and compliance (see 6.6).
k. Requirements for contractor certification (see 6.7).
l. Requirements for documentation (see 6.8).

6.3 GFP. The contracting officer should arrange to furnish the property listed in 3.7.9, 3.7.17, and 3.7.19 (see 6.2).

6.4 Metric products. Products manufactured to inch-pound dimensions will be considered on an equal basis with those manufactured using metric units, provided they fall within the tolerances specified using conversion factors in accordance with NIST SP 1038, and all other requirements of this specification are met.

6.5 Shelf-life. This specification covers items where the assignment of a Federal shelf-life code is a consideration. Specific shelf-life requirements should be as specified (see 6.2), and should include, as a minimum, shelf-life code, shelf-life package markings in accordance with MIL-STD-129 or FED-STD-123, preparation of a materiel quality storage standard for type II (extendible) shelf-life items, and a minimum of 85 percent shelf-life remaining at time of receipt by the Government. These and other requirements, if necessary, are in DoD 4140.27-M, Shelf-life Management Manual. The shelf-life codes are in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from DoD 4140.27-M, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points that manage the item and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program. Appropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website: https://www.shelflife.hq.dla.mil/.

6.6 Responsibility for inspection and compliance. Requirements for inspection and compliance are as specified (see 6.2).



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:52Z
Check the source to verify that this is the current version before use.

6.7 Contractor certification. The requirements for contractor certification are as specified (see 6.2). The Contractor should certify and maintain substantiating evidence that the product offered meets the characteristics of Section 3 of the specification and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as specified (see 6.2).

6.8 Documentation. The requirements for documentation are as specified (see 6.2). Each life raft should be shipped with a certification documentation, which includes the below-listed information, in tabular form/columns or in a spreadsheet format:

a. Contract Number
b. Life Raft Serial Number
c. Date of Manufacture
d. Date of Certification
e. Flare Lot Number
f. MROD Serial Number
g. Primary Inflation Cylinder Manufacturer and Serial Number
h. Primary Inflation Valve Manufacturer and Serial Number
i. Secondary Inflation Cylinder Manufacturer and Serial Number
j. Secondary Inflation Valve Manufacturer and Serial Number

6.8.1 Periodic reports. The Contractor should provide periodic reports, quarterly at a minimum, with life raft certification data as listed in 6.8, for new life rafts, to Logistics Engineering Resource Branch, Code 8326, Naval Surface Warfare Center Carderock Division, Detachment Norfolk (NSWCCD DN), 2600 Tarawa Ct, Suite 300, Virginia Beach, VA 23459.

6.8.2 Records management for test documentation. The Contractor should retain on file, for a period of five years, records which contain material test results (production lot test) and test records.

6.9 Additional packaging considerations for contract application. The following additional packaging considerations are as specified (see 6.2).

6.9.1 Packing. Each life raft should be packed in an enclosed wood or cardboard box crate acceptable to the common carrier, which will ensure safe delivery to destination in a satisfactory condition. Containers, packing, and method of shipment should comply with Uniform Freight Classification or National Motor Freight Classification Rules and Regulations or other carrier rules as applicable to the mode of transportation. Method of packing should facilitate handling and storage by fork lift trucks and should be capable of multiple handling and storage under favorable conditions for a minimum of 1 year. Crate or box should have tamper seal(s) on them to indicate whether or not the packaging has been opened.

6.9.2 Marking. Interior packages and exterior shipping containers, including bar coding, should be marked to identify items being delivered. The life raft, and any other included item with shelf-life requirements, should include the date of manufacturer (month and year). The shipping container should be marked with the month and year of shipment and the life raft serial number.

6.9.3 Special marking. Filled cylinders, interior containers housing same, and the exterior shipping container should contain a green label that conforms to 49 CFR 170 Subpart 179-199. In addition, the above should be marked "NON-FLAMMABLE COMPRESSED GAS" or "COMPRESSED AIR". The size and color of the markings should conform to DOT Rules and Regulations as specified herein. The markings should be conspicuous and legible.

6.9.4 Hazardous material marking. Interior packages and items, including each life raft container and exterior shipping container, should be marked and labelled to identify the compressed air or gas signal device kit, or any other hazardous material contained with the packed life raft. Markings and labels should conform to 49 CFR 171-178.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:52Z
Check the source to verify that this is the current version before use.

6.9.5 Shipping crate markings. Minimum markings should include the name and address of consignee and consignor and any applicable and any precautionary markings. Contract data markings should be placed below the identification markings.

6.9.6 Methods of marking. Marking should be accomplished by use of labels, stamping, stenciling, machine printing, or tagging. Lithographing, silk screening, photo marking, embossing, decals, transfers and bar coding of laser marking may also be used. Hand lettering or hand printing should not be used.

a. Size of lettering. Lettering for all markings should be capital letters of equal height as specified herein, so as to be clearly legible and proportional to the available marking space of the container. Unless otherwise specified herein, stenciled lettering for identification, contract data, and special markings should be not less than 6 millimeters nor more than 25 millimeters height. Address, identification, and contract data markings on plastic, fabric, paper tags, or paper labels should not be less than 2.5 millimeters.

b. Labels. Labels should be of a water-resistant grade of paper, film, fabric, or plastic, flat-back, coated on one side with pressure-sensitive adhesive. The texture of the material should be such as to permit flexibility. Labels should have a finish capable of withstanding normal handling during shipment and storage. Labels should have a finish that allows for printing and writing on with ink without feathering or spreading. The adhesive should be a pressure-sensitive permanent type. It should be homogeneous and should be coated in a smooth layer on the unprinted side of the label. The adhesive should require no solvent, heat, or other preparation prior to application. The adhesive should be of a type that will adhere to metal, plastic, or fiberboard surfaces under high or low temperatures. The applied label should remain securely in position under anticipated conditions of handling, shipment, and storage. Labels should be fade resistant.

6.10 First article inspection. When a first article inspection is required (see 6.2), the contracting officer will include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and report, and disposition of first articles. Invitations for bids will provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, will furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.11 Repair and rework. Where repair and rework is authorized by the procuring activity (see 6.2), the following guidelines should be specified:

6.11.1 Sealants. A multiplicity of pin holes in any particular section is not considered repairable. Treating or swashing the inflatable members of the life raft with latex or any other sealants to remedy pin holes is not permissible. Sealants are not permitted to remedy defects or repair damage. Sealants are permitted as part of the normal manufacturing process to enhance air retention by the coating of fabric edges at seams and cover/doubler patches to reduce wicking of air through the base fabric. Sealants are materials such as polyurethane, silicone, neoprene, etc. used to reduce wicking of air through fabric substrate.

6.11.2 Damage to base fabric. For damage incidental to the manufacturing and handling processes of the life raft where the base fabric has been penetrated, patch type repairs are permitted subject to the following:

a. Damage is not longer than 25 millimeters in any direction.

b. Damage does not enter an air retaining seam and is not within 75 millimeters of an air retaining seam.

c. Damage does not pass under or within 100 millimeters of any major anchorage point such as a towing patch, sea painter patch, or righting ladder patch.

d. There are no more than two patch type repairs for fabric damage per life raft.

e. The patch material is coated with the same type of coating as the material being repaired.

f. Any adhesive used is compatible with the coating on both the patch and repaired fabric.

g. The patch material is coated on both sides of its base fabric.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:52Z
Check the source to verify that this is the current version before use.

h. Patches extend 25 millimeters past the outer extent of the damage to be repaired and the patch corners are rounded.

i. If the patch repairs an air retaining chamber, an air retention test in accordance with 4.7.1 is successfully performed.

6.11.3 Damage to fabric coating. For incidental defects or damage of fabric coating where the base fabric has sustained no damage, patch type repairs are permitted subject to the following:

a. Damage is not longer than 25 millimeters in any direction.

b. There are no more than four patch type repairs for fabric coating damage per life raft.

c. The patch material is coated with the same type of coating as the material being repaired.

d. Any adhesive used is compatible with the coating on both patch and repaired fabric.

e. The patch material is coated on both sides of its base fabric.

f. Patches extend 25 millimeters past the outer extent of the damage to be repaired and the patch corners are rounded.

g. If the patch repairs are in an air retaining chamber, an air retention test in accordance with 4.7.1 is successfully performed.

6.12 Part numbers. Contact information for the part numbers specified herein is as follows in 6.12.1 through 6.12.6.

6.12.1 Exterior canopy lamp:

Vendor Name: Adventure Lights Inc.
CAGE Code: N/A
Web Address: www.adventurelights.com
Phone Number: (514) 694-8477
Address:
444 Beaconsfield Blvd
Beaconsfield, QC, Canada H9W 4C1

6.12.2 Interior canopy lamp:

Vendor Name: Adventure Lights Inc.
CAGE Code: N/A
Web Address: www.adventurelights.com
Phone Number: 514.694.8477
Address:
444 Beaconsfield Blvd
Beaconsfield, QC, Canada H9W 4C1

6.12.3 Sea painter system:

Vendor Name: Hampton Rubber Co. Inc.
CAGE Code: 7D712
Web Address: https://hamptonrubber.com
Phone Number: 757.722.9818
Address:
1669 W. Pembroke Ave.
Hampton, VA 23661

6.12.4 First Aid Kit (FAK):

Vendor Name: Tactical Medical (TacMed) Solutions
CAGE Code: N/A
Web Address: https://tacmedsolutions.com
Phone Number: 864.224.0081
Address:
1250 Harris Bridge Road
Anderson, SC 29621



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:52Z
Check the source to verify that this is the current version before use.

6.12.5 Towing/mooring connection hooks:

Vendor Name: Wichard America
CAGE Code: N/A
Web Address: https://www.wichardamerica.com/contact-us/
Phone Number: 401.683.5055
Address:
3901 Pine Grove Circle
Charlotte, NC 28206

6.12.6 Repair/sealing clamp kits (MIL-R-52255 & MS27826):

Vendor Name: Bridgestone Americas Tire Operations, LLC General Tires & Service
CAGE Code: 22337
Web Address: https://www.bridgestoneamericas.com
Phone Number: 615.937.3794
Address:
1200 4th Ave S, Suite 100
Nashville, TN 37201-2256

6.13 Subject term (key word) listing.

Abandon ship

Lifesaving

Survival

6.14 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

## CONCLUDING MATERIAL

Custodians:
Army – AV
Navy – SH
Air Force – 03

Review activities:
Army – MI
Navy – AS
Air Force – 11
DLA – CQ

Civil agency:
GSA – FAS

Preparing activity:
Navy − SH
(Project 4220-2020-003)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:52Z
Check the source to verify that this is the current version before use.